

SWAPCELL · DESIGN DECISIONS REGISTER

SwapCell design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/swapcell		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all open decisions 1 to 13 on 2026-10-02 (SWC-DDR-003 accepted; interface v0.4 handle zone and catch geometry; polycarbonate sheet lid; LFP code reserved; EnergyBus deferred); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the open items of SWC-DDR-001 to 003, the 2026-09-26 render review and the build plan work

SwapCell design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note (`docs/REVIEW.md`); this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The power and signal contact parts, their diameters and lengths	They set the hole sizes in the plug and receptacle shrouds and the mating order; R8 and R10 depend on them	SWC-DDR-002, SWC-DDR-003
2	The cell model: diameter 21.0 to 21.2 mm, datasheet cycle life	The holder pockets are 21.2 mm; R9 stays at risk until a cell is named	SWC-DDR-001, SWC-CAL-001
3	The battery management board's size (about 230 x 58 mm, 10 mm or less thick) and mounting holes	They set the four holes in the right side wall	SWC-DDR-003
4	The wake button's depth behind the panel, 15 mm or less	The cells are 2.7 mm behind the modelled switch body	SWC-DDR-003
5	The charger's size (modelled 150 x 64 x 110 mm)	The straps and their tapped holes are sized to it	SWC-DDR-003
6	The controller box size (modelled 50 x 50 x 30 mm)	It must fit between the charger and the shelf retainer	SWC-DDR-003
7	M3 press-in nuts rated for 1.5 mm 5052 aluminium	The lid's eight screws go into them	SWC-DDR-003

■ Value engineering

Value-engineering target: USD 700 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 606 (USD 94 under the target): USD 436 for the pack and USD 170 for the dock. Main cost drivers and savings worth trying:

- The largest lines are the 26 cells (USD 143), the battery management board (USD 120), the charger (USD 60), the plug and receptacle with their contacts (USD 60 together), the dock cradle (USD 40) and the fuses and wiring (USD 36).
- Making the design constructable added USD 47 (SWC-DDR-003): the latch housing, slider and springs (USD 10), the cell holder frames (USD 6, moved from the wiring line, which fell by USD 4), the dock mounting parts (USD 15), the longer back plate (USD 10), the tray flanges (USD 5) and more fixings (USD 5).

- Savings worth trying: a 6 mm back plate instead of 12 mm (about USD 10 and 1.5 kg less; the loads are small); cells and boards bought for several packs at once; printing the shelf and guides at lower infill; a stock enclosure-style charger with mounting ears, which would drop the straps.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 10: 13S2P of 5 Ah 21700 cells; aluminium tray with a printed flame-retardant lid; interface v0.2; SwapCell message profile with an EnergyBus gateway study; legacy discharge-only mode at 15 A; health log in the pack; dock switches a certified 5 A charger; 4.1 V fleet mode; interface governance in this repo; <code>budget_usd</code> kept at 700	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	SWC-DDR-001
2026-09-25	Interface v0.3 additions: wake for hosts without CAN (W), charge-while-discharging mode (C), latch vibration rating for vehicles (V); shared packs priced once, here	Amish, cross-cutting approval	SWC-DDR-001, items 11 to 13
2026-09-25	Connector family (custom keyed shroud with commercial contacts); values inside W, C and V confirmed (10 kΩ coding, 100 µA sleep, 25 g shock, 330 N preload, charge-FET fallback); R3 kept at 20 A with temperature derating; build budget for a second pack set before any build	Amish: "i accept all your recommendations, go with them across all repos."	SWC-DDR-002
2026-09-25	TRL 4 on hold for every repo	Amish: "Make sure we don't proceed to TRL 4 on any of them."	SWC-DDR-001, <code>docs/REVIEW.md</code>
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SWC-DDR-003 (accepted on 2026-10-02, below)
2026-09-30	Open decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	SWC-CAL-001 v0.3, this register
2026-10-02	Design for construction accepted: the changes P1 to P12 of SWC-DDR-003 and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-003, Tables 1 and 2
2026-10-02	Handle zone (option a): widened in interface v0.4 to 84 x 43 mm, from the zone's front edge to the back face, so receivers leave the thumb button clear	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-003, A1

Date	Decision	Decided by	Record
2026-10-02	Catch geometry (option a): published in interface v0.4 (latching face 1 mm above the pawl, 8 mm reach, 4 mm engagement, 6 mm pawl projection, 5 mm travel), so every receiver latches the same way	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-003, A2
2026-10-02	Lid (option b) for the prototype: cut from 3 mm UL 94 V-0 polycarbonate sheet. This changes the printed-lid decision of 2026-09-25 (SWC-DDR-001, item 2) for the prototype; the drawing allows either	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-003, A3; SWC-DDR-001
2026-10-02	Thermal path (option a): the 0.10 K/W core-to-tray assumption is kept and checked in the TRL 4 load test, with the battery board's temperature sensor placed on the rear row of cells, now furthest from the tray	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-003, A4
2026-10-02	First co-design partner: kept open under the portfolio rule, but an urban cargo-bike or e-bike delivery fleet is approached first, with a repair workshop second	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-001, item 14
2026-10-02	LFP variant: not added now; a chemistry code in the message set and a coding key are reserved so an LFP pack can never take an NMC charge or the reverse	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-001, item 16
2026-10-02	EnergyBus gateway: deferred; the native SwapCell profile stays the reference, and CiA 454 is read only when a partner needs EnergyBus	Amish: "i approve your recommendations for all 555 open decisions."	SWC-DDR-001, item 18
2026-10-02	Render pose: the lifted pose is kept for the hero render	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 1
2026-10-02	State-of-charge light bar accepted as an optional pack feature under line 14, outside the interface	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 2
2026-10-02	Dock status light accepted under line 11	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 3
2026-10-02	Thumb release behind the handle accepted; it follows from decision 2 above	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 4
2026-10-02			docs/ REVIEW.md ,

Date	Decision	Decided by	Record
	Finish and colours (slate tray, light grey lid, graphite handle, teal accents) accepted as the reference look	Amish: "i approve your recommendations for all 555 open decisions."	2026-09-26, item 5